Filippo Cenacchi

Doctoral Researcher (Computer Science) & Product Manager
Linked-in profile - Academic profile

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Profile

Doctoral Researcher in Computer Science at Macquarie University focused on cutting-edge research in humanoid AI, multimodal learning, and trustworthy healthcare intelligence. My work advances intelligent systems that can perceive behavioural signals, reason under uncertainty, interact adaptively with users, and generate auditable, human-centered decision support for preventive and clinical settings. My core strengths lie in multimodal AI, reliable machine learning, human-AI interaction, simulation-driven evaluation, and the design of deployable research systems that bridge technical innovation with real-world clinical relevance. I develop end-to-end pipelines spanning problem formulation, data acquisition, signal modelling, uncertainty-aware inference, validation, and interpretable reporting. In parallel, I bring strong product and execution capability: transforming complex stakeholder needs into measurable outcomes, scalable data workflows, and decision-ready intelligence systems. This combination of frontier research expertise and delivery strength allows me to lead technically ambitious projects from concept to rigorous, usable impact.

Teaching Experience

- **COMP8851 – Data Science and AI Major Project (2025 - ongoing)**: Teaching support for postgraduate internal-project and internship students; assessment support for project planning, progress reporting, final reports, presentations, ethical ICT practice, and professional project execution.
- **COMP8410 – Advanced Topics in Artificial Intelligence (2026 - ongoing)**: Teaching support for research-led AI topics; lab supervision, assessment feedback, student support, and applied discussion of emerging AI methods, generative models, learning systems, and evaluation.
- **COMP3850 – PACE: Computing Industry Project (2025 - ongoing)**: Teaching and marking support for work-integrated computing projects; feedback on team project planning, requirements, stakeholder communication, reports, presentations, and sponsor-facing deliverables.
- **COMP3310 – Secure Applications Development (2025 - ongoing)**: Teaching and marking support for secure software engineering; feedback on security requirements, threat modelling, implementation, testing, deployment, maintenance, and risk-aware development practice.
- **COMP1000 – Introduction to Computer Programming (2025 - ongoing)**: Teaching and marking support for introductory programming; feedback on algorithms, debugging, incremental development, practical coding tasks, and foundational computational thinking.

Education

- **Macquarie University** — Sydney, AU
  PhD, Computer Science — *2025 – 2028*
  - **Thesis Direction**: Humanoid-driven early disease detection; multimodal analysis; human-centered and ethical interactions.
- **Université Paris-Saclay** — Paris, FR
  MSc, Situated Interactions (Excellent with Honors) — *2023 – 2025*
  - **Focus**: Situated interaction design combining human, technical, and organizational constraints.
- **KTH Royal Institute of Technology** — Stockholm, SE
  MSc, Human-Computer Interactions (Excellent) — *2022 – 2024*
  - **Focus**: Human-centered design, HCI methods, and applied AI/innovation.

Teaching, Curriculum & Student Support Strengths

- **Software Engineering Education**: Able to teach across programming, requirements analysis, software design, secure development, testing, project management, professional practice, human-centred design, AI-enabled software systems, and industry project delivery.
- **Curriculum Design & Assessment**: Experience supporting authentic assessment through coding tasks, security reviews, project reports, milestone submissions, presentations, peer review, reflective writing, and sponsor-oriented deliverables. Strong focus on constructive alignment between learning outcomes, practical activities, assessment rubrics, and professional competencies.
- **Inclusive Teaching Practice**: Committed to supporting students from diverse academic, cultural, and technical backgrounds through clear explanations, structured feedback, approachable communication, scaffolded expectations, and practical examples that make complex computing concepts accessible.
- **Educational Technology**: Confident using online and blended learning tools, GitHub-based workflows, Jupyter/Colab environments, learning management systems, digital feedback workflows, collaborative whiteboarding tools, dashboards, and AI-supported learning resources to enhance engagement and learning quality.
- **Student Feedback & Mentoring**: Experienced in giving detailed, actionable feedback on code, reports, presentations, security reasoning, project documentation, teamwork, and professional communication. Strong interest in helping students build confidence, independence, and employability-oriented technical judgement.

Skills Summary

- **Programming Languages**: Python, SQL, Java, MATLAB, JavaScript, HTML, CSS.
- **Data Analysis & Visualization Tools**: Excel, Power BI, Tableau, SPSS, NVivo.
- **Product & Project Management Tools**: Jira, Confluence, Trello, Asana, Intercom, Pendo, Productboard, Miro.
- **Design, Research & Prototyping Tools**: Figma, Qualtrics, Webflow, PowerPoint.
- **Software Development & Workflow**: Git, GitHub, REST APIs, JSON, Jupyter Notebook, Google Colab, VS Code, PyCharm, Docker, Linux/Unix environments, Power BI, Tableau, and Excel-based dashboard/reporting workflows.
- **Research & Technical Applications**: PyTorch, TensorFlow, scikit-learn, XGBoost, TabPFN, SHAP, OpenCV, MediaPipe, Hugging Face Transformers, MetaHuman, Ameca/Tritium, Qualtrics, NVivo, and SPSS for multimodal AI, simulation systems, explainable machine learning, and stakeholder-facing decision-support applications.

Supervision & Leadership

- **Graduate Research Student Representative**: Faculty Research Training Committee, Macquarie University (Jul 2025 – Present): representing doctoral and master's research students in matters of training quality, policy, and student support.
- **Summer Internship Supervisor**: Macquarie University (Oct 2025 – Present): supervising summer interns in humanoid AI, multimodal fusion, and simulation-based training (Students: Chieri Ishikawa, Tuong Vy Vu).

Recent Publications

- **The Decision Twin: Metaverse Patient Digital Twins as Executable Clinical Reality**: *KDD 2026*.
- **Virtual Reality and Work: Ethical and Inclusion Implications of Facial Representation in VR**: *Journal of Strategic Information Systems*, 2025, 34(4), 101937, 1–22.
- **Personalised AI Coaching Technology: A Systematic Mapping Review**: *ACM Transactions on Interactive Intelligent Systems*, June 2026. DOI: 10.1145/3820900.
- **Passive Dementia Screening via Facial Temporal Micro-Dynamics Analysis of In-the-Wild Talking-Head Video**: arXiv:2511.13802; under review at ECCV, 2025.
- **SimClinician: An Auditable Multimodal Interface for Psychologist–AI Collaboration in Mental Health Diagnosis**: arXiv preprint; under review at *HEALTH*.
- **Learning When to Ask: Social-Timing Control for Humanoid Mental-Health Screening**: arXiv preprint; invited for rebuttal at *Robotics: Science and Systems (RSS)*, 2026.
- **Tri-Modal Severity Fused Diagnosis across Depression and PTSD**: arXiv preprint; under review at *IEEE Transactions on Affective Computing*, 2026.
- **Personalised AI Coaching Technology: A Systematic Mapping Review**: *ACM Transactions on Interactive Intelligent Systems*, 2026.
- **To Be or Not To Be: Respecting when Socially Interactive Agents Should or Should Not Conform to Expectations**: Presented at *Australasian Computer Science Week (ACSW) 2026*.
- **Beyond the Screen: Leveraging Virtual Reality and Socially Interactive Agents to Redefine Bias Mitigation**: Under review at *Journal of Multimodal User Interfaces*, 2026.
- **Perceptions of Facial Misrepresentation in Virtual Reality: An Ethical Analysis**: Under revision at *Ethics and Information Technology*, 2025.
- **Humanoid AI Robots in Early Disease Detection: Challenges, Techniques and Opportunities**: Under review at *ACM Computing Surveys*.
- **FALCON-Discover: Discovering Concentrated False-Confidence Regions for Calibration**: Submitted to *NeurIPS 2026*.
- **Beyond Output-Space Calibration: Spectral Evidence Bundling for Selective Reliability Estimation in Time-Series Classification**: Submitted to *NeurIPS 2026*.
- **Counterfactual Fragility Certificates: Exposing High-Confidence Brittleness under Structured Evidence Failure**: Submitted to *NeurIPS 2026*.
- **Understanding Low-Code/No-Code Adoption in Software Startups**: In *PROFES 2022*.

Invited Talks & Keynotes

- **KDD Blue Sky Keynote: The Decision Twin**: Keynote presentation at *KDD 2026 Blue Sky Track*, Jeju, South Korea, 2026.
- **Can humanoid robots help diagnose cognitive and affective health?**: Invited talk at School of Computing Industry Networking Event, Macquarie University, 21 Nov 2025.

Recent Industry Experience (See full list)

- **Meta** — Sydney, AU
  Researcher (Contract) — *Feb 2024 – Jan 2025*

  Ethical VR research in Macquarie partnership; user-to-user communication studies; recommendations for safer and more inclusive virtual interaction design.

Industry & Software Engineering Practice

- **Product Management & Software Delivery**: Product management experience across SaaS, research, and AI-enabled software environments, including translating user needs, stakeholder requirements, technical constraints, and business goals into product decisions, implementation priorities, roadmaps, and measurable delivery outcomes. At GenValues AB, contributed to product and customer-facing workflows involving platform improvement, stakeholder feedback, feature prioritisation, and data-informed product decisions.
- **Agile & Professional Practice**: Practical experience with agile and cross-functional software delivery practices, including backlog refinement, sprint planning, roadmap execution, user research, requirements analysis, stakeholder communication, and evidence-based prioritisation. Experienced with Jira, Confluence, Trello, Asana, Intercom, Pendo, Productboard, Miro, GitHub, REST APIs, dashboards, and collaborative documentation workflows.
- **Applied Software Engineering Data Workflows**: Developed and supported data-driven workflows using Python, SQL, Excel, Power BI, Tableau, Jupyter Notebook, Google Colab, Git/GitHub, Docker, Linux environments, JSON-based data exchange, and API-oriented workflows. Experience spans research data processing, dashboarding, model evaluation, reporting pipelines, and stakeholder-facing analytics for decision support.
- **AI, GPU & Modern Computing Practice**: Hands-on experience with modern AI/software stacks including PyTorch, TensorFlow, scikit-learn, XGBoost, Hugging Face Transformers, OpenCV, MediaPipe, SHAP, and GPU-enabled model experimentation using NVIDIA/CUDA-compatible workflows where available. Able to connect software engineering teaching with contemporary industry practice in AI systems, data pipelines, reproducible experimentation, deployment constraints, and responsible evaluation.
- **Industry-Relevant Software Engineering Examples**: Able to teach software engineering through authentic examples from product management, AI research, VR/AR systems, human-computer interaction, and healthcare technology, including requirements elicitation, risk analysis, usability evaluation, stakeholder negotiation, version control, testing, documentation, deployment planning, and ethical technology design.
- **Work-Integrated Learning Relevance**: Experience supporting work-integrated learning through COMP3850 PACE and postgraduate project teaching, helping students connect classroom learning with sponsor-facing deliverables, requirements analysis, project planning, client communication, risk management, reflective practice, professional reporting, and presentation of technical outcomes to non-specialist stakeholders.
- **Human-Centred and Responsible Technology**: Strong background in HCI, VR, AI, ethics, inclusion, and user-centred evaluation through research with Meta/Macquarie University and doctoral work on multimodal and humanoid AI. Able to teach students how software engineering decisions affect users, organisations, accessibility, safety, fairness, privacy, accountability, and real-world technology adoption.

Prizes & Grants

- **Macquarie University Research Excellence Scholarship iMQRES(ID:20246179)**: $117k scholarship, 2025.
- **Macquarie University 3MT**: School of Computing 3MT: 2nd Place & People's Choice (Prize: $300), 2025.
- **Macquarie University 3MT**: Faculty of Science & Engineering 3MT: 2nd Runner Up (Prize: $250), 2025.
- **Macquarie University 3MT**: University 3MT: 2nd Runner Up (Prize: $350), 2025.
- **PERC Best Research Award**: Recognition at Performance and Expertise Research Centre Conference (Macquarie University), Sydney, 2025.

Licenses & Certifications

- **Reviewer, IJCAI 2025 & 2026 (International Joint Conferences on Artificial Intelligence Organization)**:
- **Strategising: Management for Global Competitive Advantage (Specialization) (Macquarie University)**: Issued Jul 2024 Credential ID: 3SBZUD34JHWU
- **Developing an Agile Team (University of Colorado)**: Issued Jun 2024 Credential ID: ARHLT65CD8R2
- **Budgeting and Scheduling Projects (University of California, Irvine (Paul Merage School of Business))**: Issued Jun 2024Credential ID: JVVPJKEX5W7P
- **Storytelling and Influencing: Communicate with Impact (Macquarie University)**: Issued May 2024 Credential ID: 6B8QRC3ZKTMT
- **Advanced Competitive Strategy (Ludwig-Maximilians-Universität München)**: Issued May 2024 Credential ID: ZMCJ78UUU9BM
- **Excel Fundamentals for Data Analysis (Macquarie University)**: Issued Apr 2024 Credential ID: 353FUA3QNZLU
- **Nova Membership Certificate (Top 3% Talent Among 2024 Graduates in Italy)**: Nova, Issued Oct 2022 Credential ID: 2W8KBSXMO66BV78ZNSI4ZG3KJ
- **IELTS Academic Certificate C2 (IELTS Official)**: Credential ID: A3-IT010-S-12590198